

SIMATS ENGINEERING  
ASSESSMENT TOOL 2  
Case study

COURSE NAME: Computer Networks

COURSE CODE:  
CSA07

COURSE OUTCOME COVERED

**CO3:** *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4  
Marks:40

Total

1. A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an 8% packet loss, 180 ms jitter, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.

Problem Analysis

The conferencing problem is mainly caused by **network congestion and the handling of real-time traffic at the transport and application layers**. The network shows **8% packet loss, 180 ms jitter, and fluctuating bandwidth**, which directly results in distorted audio, frozen video, and delayed communication.

Real-time conferencing commonly uses **UDP** because it provides faster delivery without waiting for retransmissions. However, UDP does not guarantee packet delivery, so high packet loss can still affect communication. The application layer can also contribute if its buffering and error-handling mechanisms are not properly adapted to changing network conditions.

Key Issues

- **8% packet loss** causes missing audio and video information.
- **180 ms jitter** causes packets to arrive at inconsistent times.
- **Bandwidth fluctuation** causes congestion and reduced media quality.
- Poor **buffering strategies** can increase delay.
- UDP is suitable for real-time traffic but does not recover lost packets automatically.

2. Justify your answer based on the communication requirements of real-time

## **multimedia applications.**

### **Need for Low Delay**

Video conferencing requires packets to arrive **quickly and continuously**. Unlike file transfers, delayed packets are often useless because the corresponding audio or video has already been played.

### **UDP vs TCP**

**UDP** is generally preferred for real-time multimedia because it avoids the retransmission and acknowledgement delays associated with TCP. TCP guarantees reliable and ordered delivery, but retransmissions can introduce additional latency and head-of-line blocking.

### **Important Requirements**

Real-time multimedia applications require:

- **Low latency** for natural conversations.
- **Low jitter** for smooth audio and video.
- **Low packet loss** to avoid missing information.
- **Stable bandwidth** for continuous media transmission.
- **Adaptive buffering** to handle network variations.

Therefore, the observed **8% packet loss and 180 ms jitter** are major contributors to the poor conferencing quality.

### **3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.**

#### **Protocol and Network Improvements**

The network should prioritize real-time voice and video traffic using **Quality of Service (QoS)** mechanisms. UDP-based real-time media transport should be used where appropriate, while congestion and bandwidth usage should be controlled.

- Implement **QoS** to prioritize voice and video.
- Reduce network congestion during peak hours.
- Allocate sufficient bandwidth for conferencing.
- Use suitable real-time protocols such as **RTP/RTCP**.
- Monitor packet loss, jitter, latency, and bandwidth continuously.

#### **Application-Level Improvements**

The conferencing application should dynamically adapt to changing network conditions. An **adaptive jitter buffer** can temporarily store packets to reduce the effects of variable packet arrival times.

- Use **adaptive jitter buffering**.
- Implement **adaptive bitrate** for video.
- Dynamically reduce video resolution during congestion.
- Use **packet-loss concealment**.
- Apply **forward error correction** where appropriate.

### **2. A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the average DNS lookup time is approximately 800 ms, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.**

## **Questions**

1. Analyze the reasons behind the high DNS resolution time.

DNS Resolution Analysis

The average DNS lookup time of **800 ms** is very high and can significantly delay the loading of the application homepage. DNS resolution requires the client to obtain the server's IP address before establishing the connection. The high delay may be caused by **geographical distance between users and DNS servers, network congestion, overloaded DNS servers, inefficient recursive lookups, and poor DNS caching.**

Major Reasons

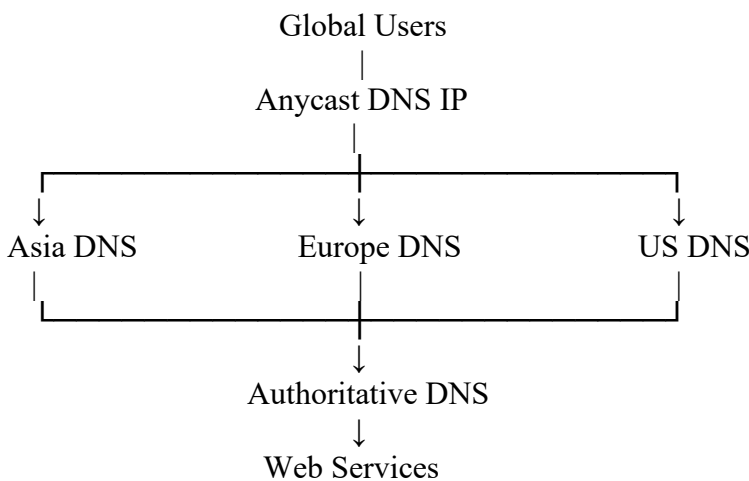
- **Geographical distance** increases the round-trip time for DNS queries.
- **Network congestion** can delay DNS requests and responses.
- **Poor caching** causes repeated DNS lookups.
- **DNS server overload** increases processing and response time.
- **Multiple recursive queries** may be required before obtaining the final IP address.
- **Low TTL values** cause cached records to expire frequently.
- **Centralized DNS infrastructure** can create a performance bottleneck.

Therefore, the 800 ms delay is mainly due to inefficient DNS resolution paths combined with inadequate caching and distributed infrastructure.

2. Propose a suitable DNS architecture using techniques such as Anycast DNS, DNS pre-fetching, and TTL optimization to reduce the lookup time below 50 ms.

Anycast DNS

The company should deploy **Anycast DNS servers across multiple continents**, such as Asia, Europe, and North America. All DNS servers can advertise the same Anycast IP address, allowing DNS requests to be routed to a nearby available server.



DNS Pre-fetching

**DNS pre-fetching** can be used to resolve frequently accessed domains before users actually request them. Frequently used DNS information can remain available in caches, reducing the time required for future lookups.

TTL Optimization

The organization should use **appropriate TTL values** for DNS records. Stable records can have longer TTL values so that recursive DNS servers keep them cached for longer periods. This

reduces repeated queries and improves response time.

### Proposed Improvements

- Deploy DNS servers in multiple geographical regions.
- Use **Anycast DNS** to direct users to nearby DNS servers.
- Enable DNS caching and pre-fetching.
- Optimize TTL values based on how frequently records change.
- Use multiple authoritative DNS servers for redundancy.
- Continuously monitor DNS latency and server availability.

This architecture can significantly reduce DNS latency and can target **less than 50 ms for many users**, provided the DNS nodes are appropriately distributed and network paths are healthy.

## 3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

### A. Performance

#### Advantages

- Anycast reduces geographical distance to DNS servers.
- DNS caching reduces repeated lookups.
- Pre-fetching speeds up frequently requested domains.
- Optimized TTL values improve cache hit rates.
- Faster DNS resolution improves initial page-loading time.

#### Limitations

- DNS performance still depends on the user's network and recursive resolver.
- Network congestion can continue to cause delays.
- Pre-fetching can generate additional DNS queries.
- Excessively long TTL values may cause stale DNS information.

### B. Scalability

#### Advantages

- Multiple DNS servers can handle large numbers of queries.
- Regional servers distribute DNS traffic.
- New DNS nodes can be added as the company expands.
- Caching reduces the load on authoritative DNS servers.

#### Limitations

- Global DNS infrastructure is more complex to manage.
- Additional servers require monitoring and maintenance.
- Anycast routing requires careful configuration.
- Infrastructure and operational costs increase.

### C. Availability

#### Advantages

- Multiple DNS locations provide **fault tolerance**.
- Failure of one DNS server does not necessarily interrupt the service.
- Anycast can route requests toward another available location.
- Multiple authoritative servers prevent dependence on a single server.

#### Limitations

- Routing failures can affect access to an Anycast location.
- Configuration errors can affect multiple DNS servers.
- Distributed infrastructure requires continuous monitoring.
- Failover mechanisms must be properly tested.

3. A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly 800 ms because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

## Questions

1. Examine the impact of centralized DNS architecture on application performance.

### DNS Performance Analysis

A centralized DNS architecture directs all DNS queries to a **single primary DNS server**. When millions of users access cloud services during promotional events, the server receives a very large number of requests. This can overload the server and increase DNS response time to nearly **800 ms**, causing delays before users can connect to the cloud-hosted application.

### Major Impacts

- **High DNS latency:** Slow DNS resolution delays application access.
- **Server overload:** A single server must process all incoming queries.
- **Single point of failure:** If the DNS server fails, users may be unable to resolve the application domain.
- **Network congestion:** Concentrating queries on one server can create traffic bottlenecks.
- **Poor scalability:** The architecture cannot easily handle sudden increases in global traffic.
- **Reduced availability:** Server or network failure can affect users worldwide.

Therefore, centralized DNS negatively affects both **application performance and service availability**, especially during high-traffic events.

2. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

### Proposed Architecture

The company should replace the single DNS server with **multiple geographically distributed DNS servers**. DNS servers can be deployed in regions such as Asia, Europe, and North America and configured to provide the same DNS service.

### Anycast Routing

**Anycast DNS** allows multiple DNS servers to advertise the same IP address. User queries are automatically routed toward an appropriate nearby DNS location, reducing geographical distance and lookup latency.

### DNS Caching

Frequently requested domain records should be cached by recursive DNS resolvers. Cached responses can be returned directly without contacting the authoritative DNS server for every request.

### High Availability

Multiple DNS servers provide redundancy. If one DNS server or geographical region becomes unavailable, DNS requests can continue to be handled by other available servers.

### Key Features

- Deploy DNS servers across multiple geographical regions.
- Use **Anycast routing** to distribute DNS requests.
- Configure multiple authoritative DNS servers.
- Enable effective DNS caching.
- Use suitable TTL values for DNS records.
- Monitor DNS servers and provide automatic failover.

This distributed architecture removes the single-server bottleneck and provides **lower lookup latency, better scalability, and higher availability**.

### 3. Analyze how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance.

#### A. TTL Values

**TTL (Time to Live)** determines how long a DNS record remains in a resolver's cache.

##### Advantages

- Reduces repeated DNS queries.
- Decreases load on authoritative DNS servers.
- Improves response time for cached requests.
- Helps the system handle large traffic volumes.

##### Limitations

- Very long TTL values can cause **stale DNS records**.
- Very short TTL values cause frequent DNS lookups.
- Frequent lookups increase server load and network traffic.

Therefore, the company should select **optimized TTL values** based on how frequently its DNS records change.

#### B. Anycast Routing

Anycast allows several DNS servers in different locations to use the **same IP address**.

##### Advantages

- Routes users to a nearby DNS server.
- Reduces DNS lookup latency.
- Distributes traffic across multiple locations.
- Improves fault tolerance.
- Provides service continuity when one DNS location fails.

##### Limitation

Anycast depends on correct routing information. Routing failures or incorrect advertisements can cause traffic to reach an unsuitable or unavailable DNS server.

#### C. DNS Caching

DNS caching stores previously resolved domain information for a specified period.

##### Advantages

- Provides faster DNS responses.
- Reduces queries reaching authoritative servers.
- Reduces network traffic.
- Improves application startup time.
- Helps handle large numbers of repeated requests.

## Limitation

Cached information may become outdated until the TTL expires.

4. A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the average order acknowledgment latency increases from 1 ms to 40 ms, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

## Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering flow control, Nagle's algorithm, and SYN queue limitations.

## TCP Flow Control

TCP uses a **receive window** to control how much data can be sent before the receiver must acknowledge it. In a high-frequency trading system, a small or dynamically reduced receive window can limit the amount of data transmitted at once. If the receiver or network cannot process incoming data quickly enough, the sender may be forced to wait, increasing order acknowledgment latency.

## Nagle's Algorithm

**Nagle's algorithm** combines small TCP packets into larger segments to reduce network overhead. While useful for general applications, it can introduce delays when trading systems send many **small order and acknowledgment messages**. TCP may wait for an acknowledgment or additional data before transmitting a small packet, increasing the time required to process an order.

## SYN Queue Limitations

The **SYN queue** stores partially established TCP connections during the TCP three-way handshake. During market opening, thousands of clients may attempt to connect simultaneously. If the SYN queue becomes full, new connection requests may be delayed or dropped, resulting in longer connection establishment times and retransmitted SYN packets.

## Key Factors

- **Flow control:** Limits data transmission when the receive window is constrained.
- **Nagle's algorithm:** Can delay small trading messages.
- **SYN queue limitations:** Cause connection delays during sudden traffic spikes.

2. Explain how each factor affects transaction processing in high-frequency systems

## Flow Control Impact

High-frequency trading requires extremely fast exchange of small messages. If TCP flow control reduces the sending rate, order messages and acknowledgments may remain in buffers instead of being transmitted immediately.



**Effects:**

- Increased acknowledgment latency.
- Excessive buffering.
- Lower transaction throughput.
- Delayed order confirmations.

**Nagle's Algorithm Impact**

Trading platforms frequently exchange small packets such as **buy orders, sell orders, and acknowledgments**. Nagle's algorithm can combine these packets, but the waiting time introduced by aggregation is undesirable for latency-sensitive transactions.

**Effects:**

- Small packets may wait before transmission.
- Order acknowledgments can be delayed.
- Latency becomes more variable.
- Real-time trading performance decreases.

**SYN Queue Impact**

During market opening, many users and trading systems may establish connections simultaneously. A limited SYN queue can become saturated, causing connection requests to wait or be retransmitted.

**Effects:**

- Longer TCP connection establishment.
- Increased SYN retransmissions.
- Higher server CPU/network overhead.
- Delayed order submission.
- Possible connection failures during traffic peaks.

### **3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.**

**Disable Nagle's Algorithm**

For latency-sensitive trading connections, **TCP\_NODELAY** can be enabled to disable Nagle's algorithm. This allows small packets to be transmitted immediately instead of waiting for additional data.

**Benefit:**

- Faster order transmission.
- Faster acknowledgments.
- Reduced application-level latency.

**Optimize TCP Receive/Send Buffers**

TCP send and receive buffer sizes should be configured appropriately for the application's traffic pattern. The buffers should be large enough to prevent unnecessary blocking but not so large that excessive queuing occurs.

**Benefit:**

- Reduces unnecessary waiting.
- Improves throughput.
- Minimizes excessive buffering.

**Increase SYN Queue Capacity**



The server should increase the **SYN backlog/queue size** to handle large numbers of simultaneous connection attempts during market opening.

**Benefit:**

- Handles connection bursts more effectively.
- Reduces SYN drops.
- Decreases connection establishment delays.

**Enable SYN Cookies**

**SYN cookies** can provide protection when the SYN queue becomes overloaded by allowing the server to handle large numbers of connection attempts without relying entirely on queued connection state.

**Reduce Retransmissions**

Network engineers should investigate packet loss, congestion, and unstable network paths that are causing the observed TCP retransmissions.

**Possible actions:**

- Improve network capacity.
- Reduce packet loss.
- Monitor TCP retransmission rates.
- Optimize routing paths.
- Maintain stable low-latency connections.

RUBRICS FOR CALCULATION

| Criteria                | Weightage | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                    | Level 1 – Beginning                                  |
|-------------------------|-----------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------------|
| Understanding of Case   | 25        | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |
| Application of Concepts | 25        | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts         | Fails to apply relevant concepts                     |
| Analysis                | 20        | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth       | Weak or no analysis; lacks reasoning                 |
| Solution Design         | 20        | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification | Incorrect or no solution provided                    |
| Clarity                 | 10        | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure           | Poor clarity; difficult to understand                |